

Page Turner

Bimanual Robotic Book Page-Turning System

Presentation 2 · TECHIN 517 · June 2026

Team Members

Josh D Kirby · Model Training & Code

Alan Liu · Data Recording&

Xirui Zhu · Testing & Data Collection

Page Turner · Quantitative Evaluation Results

TECHIN 517 · May 2026

Raw Trial Data · 30 Runs across 3 Conditions

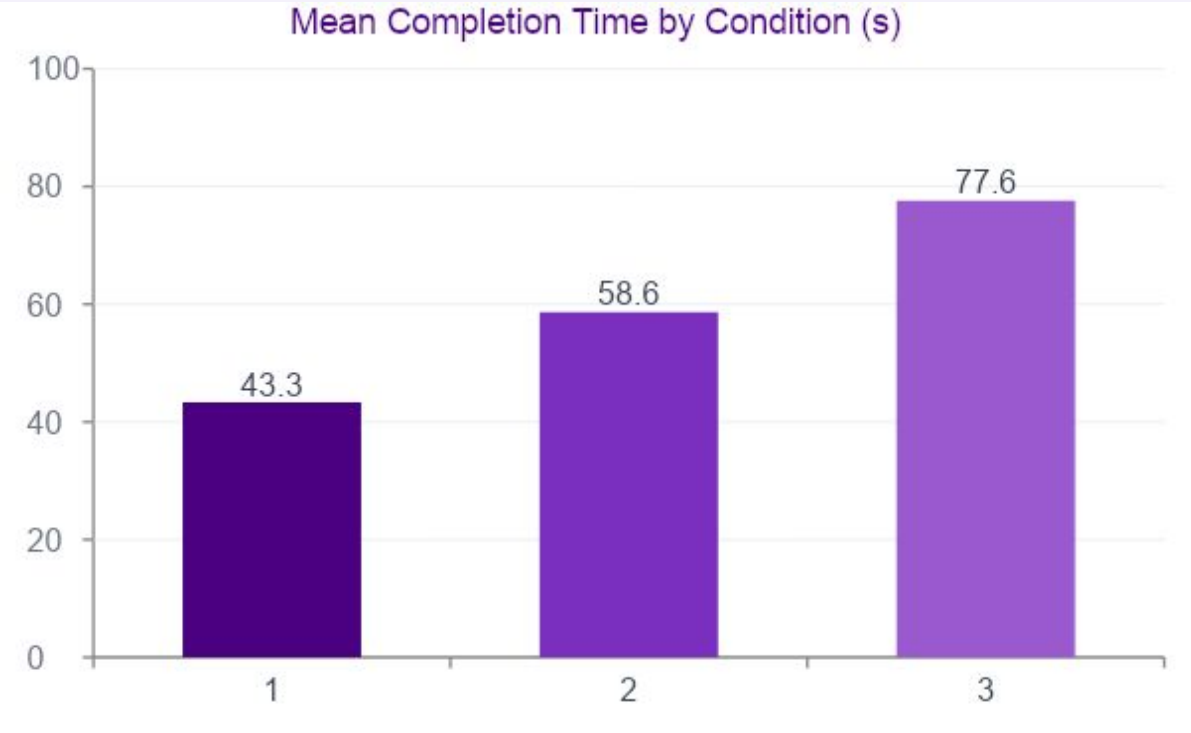
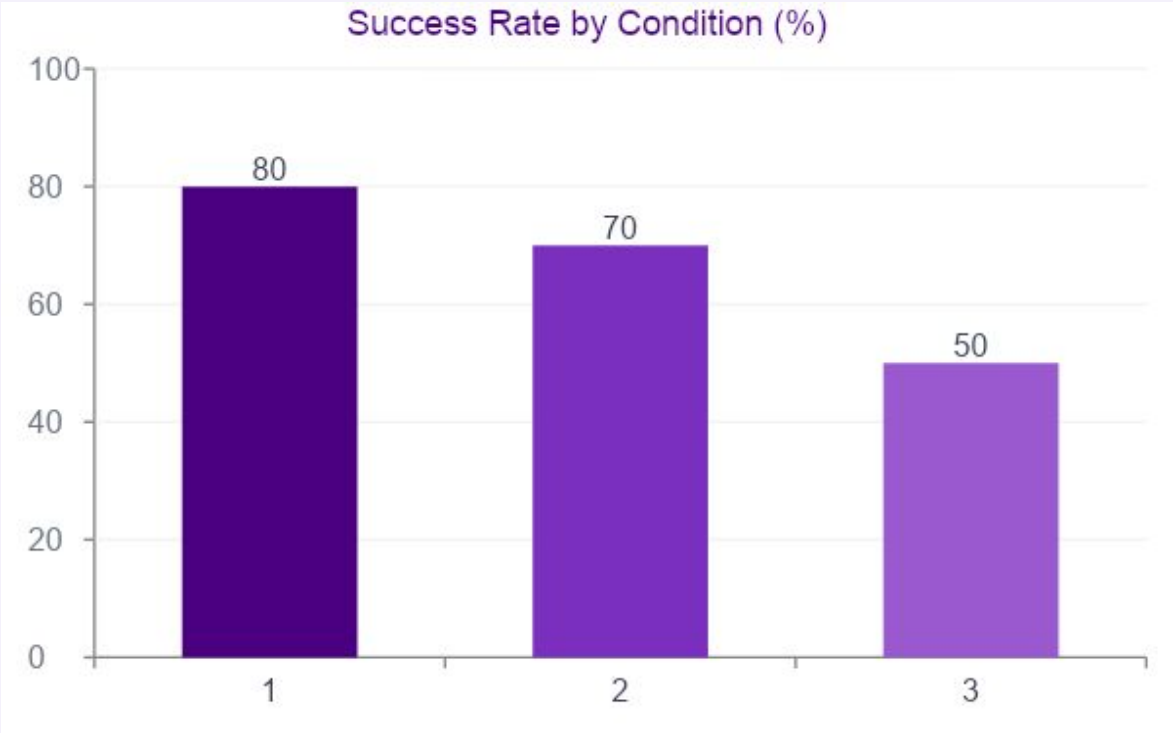
Trial	Condition	Result	Time (s)	Failure Mode	Notes
A-01	A – Training Book	✓ Pass	42	—	Placeholder observation
A-02	A – Training Book	✓ Pass	38	—	Placeholder observation
A-03	A – Training Book	✓ Pass	51	—	Placeholder observation
A-04	A – Training Book	✓ Pass	45	—	Placeholder observation
A-05	A – Training Book	✗ Fail	—	Page not turned	Placeholder observation
A-06	A – Training Book	✓ Pass	39	—	Placeholder observation
A-07	A – Training Book	✓ Pass	47	—	Placeholder observation
A-08	A – Training Book	✓ Pass	43	—	Placeholder observation
A-09	A – Training Book	✗ Fail	—	Timeout	Placeholder observation
A-10	A – Training Book	✓ Pass	41	—	Placeholder observation
B-01	B – Novel Paperback	✓ Pass	55	—	Placeholder observation
B-02	B – Novel Paperback	✓ Pass	60	—	Placeholder observation
B-03	B – Novel Paperback	✗ Fail	—	Arm stopped	Placeholder observation
B-04	B – Novel Paperback	✓ Pass	58	—	Placeholder observation
B-05	B – Novel Paperback	✓ Pass	52	—	Placeholder observation
B-06	B – Novel Paperback	✓ Pass	67	—	Placeholder observation
B-07	B – Novel Paperback	✗ Fail	—	Page not turned	Placeholder observation
B-08	B – Novel Paperback	✓ Pass	61	—	Placeholder observation
B-09	B – Novel Paperback	✓ Pass	57	—	Placeholder observation
B-10	B – Novel Paperback	✗ Fail	—	Object dropped	Placeholder observation

Trial	Condition	Result	Time (s)	Failure Mode	Notes
C-01	C – Novel Hardcover	✓ Pass	74	—	Placeholder observation
C-02	C – Novel Hardcover	✗ Fail	—	Timeout	Placeholder observation
C-03	C – Novel Hardcover	✓ Pass	81	—	Placeholder observation
C-04	C – Novel Hardcover	✗ Fail	—	Safety stop	Placeholder observation
C-05	C – Novel Hardcover	✗ Fail	—	Page not turned	Placeholder observation
C-06	C – Novel Hardcover	✓ Pass	70	—	Placeholder observation
C-07	C – Novel Hardcover	✓ Pass	78	—	Placeholder observation
C-08	C – Novel Hardcover	✗ Fail	—	Arm stopped	Placeholder observation
C-09	C – Novel Hardcover	✓ Pass	85	—	Placeholder observation
C-10	C – Novel Hardcover	✗ Fail	—	Timeout	Placeholder observation

Summary Statistics · Success Rate & Timing

Per-condition breakdown with mean completion time and failure analysis

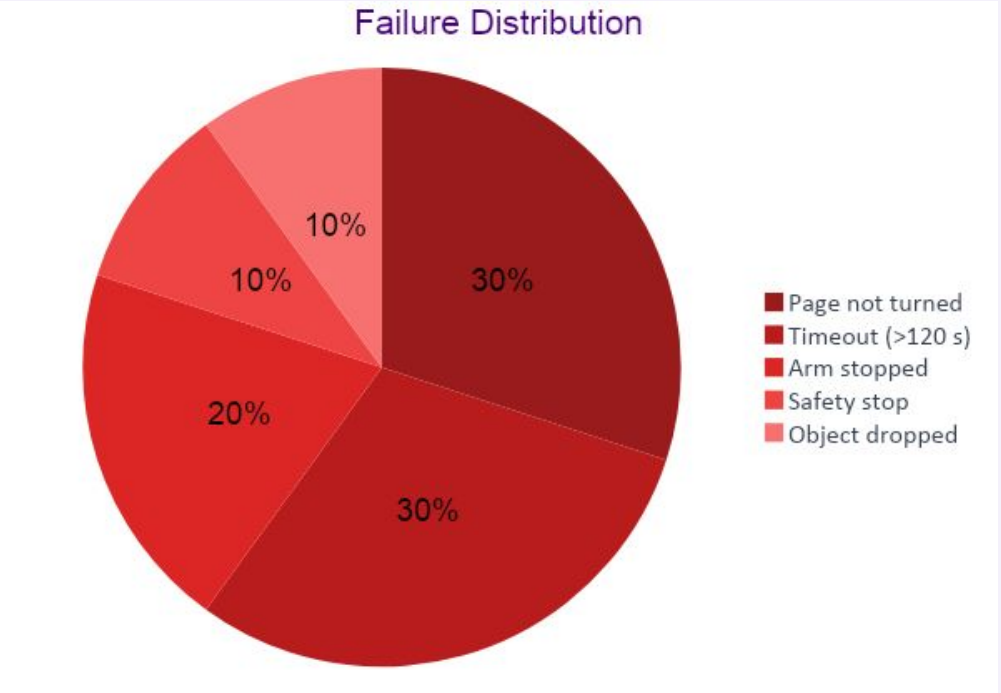
Condition	Trials	Successes	Success Rate	Mean Time (s)	Std Dev (s)
A – Training Book	10	8	80%	43.3	4.0
B – Novel Paperback	10	7	70%	58.6	4.4
C – Novel Hardcover	10	5	50%	77.6	5.2
Overall	30	20	67%	57.2	14.2



Failure Mode Analysis · Classification & Patterns

All 10 failures categorized across conditions A / B / C

Failure Mode	Cond A	Cond B	Cond C	Total	Description / Root Cause (placeholder)
Page not turned	1	1	1	3	Policy fails to arch/grip page; likely insufficient page-edge detection
Timeout (>120 s)	1	—	2	3	Task runs past limit; slow convergence on novel book geometry
Arm stopped	—	1	1	2	Safety stop or motor fault mid-trajectory
Safety stop	—	—	1	1	Collision threshold exceeded; gripper contacts spine or table
Object dropped	—	1	—	1	Page slips from gripper during flip motion

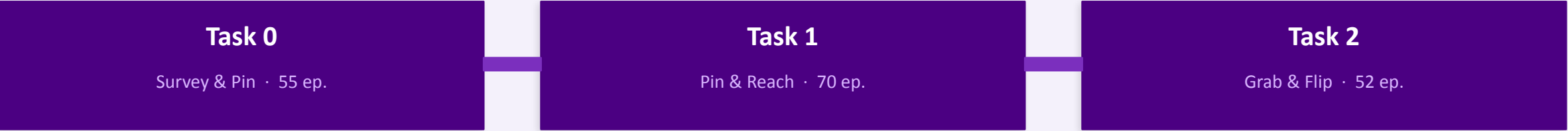


Key Observations

1. Condition C (hardcover) has the highest failure rate (5/10), primarily due to stiffer pages resisting the arch in Task 0.
2. Timeout failures increase with book novelty — longer inference time on unseen geometry.
3. Safety stops only occur on hardcover, suggesting the gripper contacts the stiff spine during the flip.
4. Condition A (training book) performs best (8/10), validating the ACT policy on the in-distribution object.
5. All failure modes are recoverable — no hardware damage observed in any trial.
6. → Next steps: augment Task 0 training data with hardcover demos; tune collision threshold for stiffer books.

End-to-End Pipeline Evaluation · Task 0 → 1 → 2

Full autonomous page-turn: no human intervention between sub-tasks · 10 trials on training book



Trial	Result	Task 0 (s)	Task 1 (s)	Task 2 (s)	Total (s)	Failure Point	Notes
E-01	✓ Pass	42	38	31	111	—	Smooth handoff; book stayed in position
E-02	✓ Pass	39	41	29	109	—	Fastest run; page arch clean in Task 0
E-03	✗ Fail	51	—	—	—	Task 1 – Arm stopped	Left gripper failed to reach around arch
E-04	✓ Pass	45	44	33	122	—	Slightly over target; Task 1 slow convergence
E-05	✗ Fail	—	—	—	—	Task 0 – Timeout	Book edge detection failed on first attempt
E-06	✓ Pass	43	39	30	112	—	Clean end-to-end; no replanning needed
E-07	✓ Pass	47	42	35	124	—	Page slightly misaligned at end; counted pass
E-08	✗ Fail	38	36	—	—	Task 2 – Page not turned	Gripper slipped during final flip
E-09	✓ Pass	44	40	32	116	—	Normal run
E-10	✓ Pass	41	43	28	112	—	Fastest flip (28 s); good page arch

7/10

Success Rate

70% end-to-end

115.1 s

Mean Total Time

± 5.4 s (std dev)

≥ 80%

Pipeline Target

< 120 s per run